# Kick-off document

## Understanding of the Project Objectives

The project aims to develop a causal model explaining Bitcoin's performance across different time horizons (day, week, month, quarter, year).
The goals can be summarized as follows:

1. Primary objective:
   Build a causal model identifying and quantifying the key drivers behind Bitcoin's performance, and analyze how these causal relationships evolve with time.
2. Secondary objective:
   Use the insights from the causal model to design a slow-moving investment strategy capable of capturing upside potential while mitigating downside risks.

This work addresses a crucial challenge for institutional investors: the need to rationalize and justify investment decisions in a transparent way, as traditional machine learning methods often lack interpretability.

## Breakdown of planned tasks (à faire

**1. Data Pipeline**

- Collect Bitcoin-related data from APIs and financial databases
- Clean, structure, and document the dataset for analysis

**2. Feature Selection**

- Perform exploratory data analysis (EDA)
- Identify and select the most relevant features using statistical and domain knowledge

**3. Causal Model Construction**

- Build a Directed Acyclic Graph (DAG) showing causal relationships
- Test causal links using tools like DoWhy or CausalNex

**4. Evaluation and Validation**

- Compare causal results with ML models (regression, random forest, LSTM)
- Assess accuracy, robustness, and interpretability

**5. Strategic Application**

- Use insights to design a slow-moving investment strategy.
- Summarize findings and finalize the Jupyter Notebook.

## Overall Vision of the Organization

The project will be conducted collaboratively, with each team member contributing according to their background and technical expertise.
The goal is to combine data science, finance, and engineering perspectives to build a robust and interpretable causal model of Bitcoin performance.

**Nicolas Cosmano (IF):** Finance, Bitcoin

**Alexandre KOCH (Fintech):** Blockchain, crypto markets

**Xu Zilong (DIA):** Exploratory data analysis, Data collection, Modeling

**Hugo Nicolay (DIA):** ML methods, python basics

**Alvaro Serero (DIA):** ML models performance, data cleaning (pandas in Python)

**Marcel Yammine(DIA):** ML methods and a good understanding of cryptocurrency markets and blockchain fundamentals.

## Detailed Planning

| Period | Key Activities | Responsible | Deliverables |
| --- | --- | --- | --- |
| **Weeks 1–2** | Kick-off, project understanding, initial data exploration | All | This kick-off document |
| **Weeks 3–4** | Data collection and cleaning | All | Validated dataset |
| **Weeks 5–6** | Causal model design and graph creation | All | Preliminary DAG |
| **Weeks 7–8** | Model testing, accuracy validation, ML comparison | All | Analytical report |
| **Weeks 9–10** | Investment strategy design and final presentation | All | Final notebook + oral presentation |

## Deliverables and Timeline

| Deliverable | Content | Expected Date |
| --- | --- | --- |
| **Kick-off Document** | Project vision, organization, planning | Week 2 |
| **Dataset & Pipeline** | Clean, structured data with documentation | Week 4 |
| **Causal Model Prototype** | First causal graph and interpretation | Week 6 |
| **Comparison Report** | Causal vs ML model analysis | Week 8 |
| **Final Deliverables** | Complete notebook, strategic insights, and presentation | Week 10 |